

Sequential decomposition and attribution for trajectory forecasting

Yonghui Liu , Inhi Kim *

Korea Advanced Institute of Science and Technology, Daejeon, Republic of Korea

ARTICLE INFO

Keywords:

Trajectory forecasting
Sequential decomposition
Input attribution
Temporal dependence
Goal-conditioned prediction
Intelligent transportation systems
Model interpretability

ABSTRACT

Trajectory forecasting is fundamental to intelligent mobility systems. A prevailing assumption is that longer historical inputs consistently help, which has motivated the design of models with extended observation windows. Yet empirical results often show diminishing returns, higher computational cost, and increased sensitivity to noise. Despite extensive architectural innovation, there is still no unified framework for systematically attributing forecasting performance to different temporal inputs. Rather than introducing a new forecasting model, we present the *Sequential Decomposition and Attribution* (SDA) protocol, a model-agnostic *evaluation paradigm* that decomposes inputs into *Past* (P), *Current* (C), and *Prior* (R, i.e., destination or goal when available) and quantifies their standalone utility, marginal contributions, and interactions through controlled input gating. Instantiated on trajectory forecasting on two real-world datasets (Porto taxi and ETH pedestrian) with two representative backbones (LSTM and Transformer), SDA shows that accuracy gains from history saturate with short sequences, the most recent state alone is highly informative at short horizons, and reliable priors yield substantial improvements with greater value at longer horizons. Interaction analyses further reveal conditional complementarity among sources. These findings highlight the importance of input-aware designs that prioritize efficient use of the latest state and goal-conditioning over indiscriminate history accumulation. While demonstrated on trajectories, SDA offers a general lens for sequential information attribution.

1. Introduction

Trajectory forecasting is a fundamental task in intelligent transportation and mobility systems. It supports a wide range of downstream applications, including taxi dispatching, route planning, pedestrian navigation, and autonomous driving [1,2]. A common modeling intuition is that longer historical input provides richer temporal context and thereby leads to more accurate predictions. This intuition has shaped the design of many forecasting models, including recurrent neural networks [3], spatiotemporal graph encoders [4], and Transformer-based architectures [5,6], where the observation window is chosen as a fixed hyperparameter.

Recent empirical evidence suggests that this intuition does not always hold. Models with greater capacity to capture long-range temporal dependencies, such as Transformers with global self-attention, do not consistently outperform simpler recurrent models in benchmark trajectory prediction tasks [7,8]. These results indicate that extending history may yield diminishing returns, while also introducing additional costs in

computation, memory, and robustness. In practice, long input windows may amplify noise or outdated signals without proportionate predictive gains.

Although many studies have proposed architectural innovations for trajectory forecasting, much less work has systematically examined how different portions of the observed trajectory affect prediction. Most models encode the input sequence as a single block of data, without distinguishing the relative value of early history, the latest state, or destination-related information. As a result, the marginal roles of these temporal and exogenous sources remain underexplored in a consistent manner, limiting our understanding of what kinds of information are most valuable.

To address this gap, we propose the *Sequential Decomposition and Attribution* (SDA) protocol, a model-agnostic framework that decomposes trajectory inputs into three sources: the *Past* (historical observations), the *Current* (the most recent observed state), and the *Prior* (goal or destination information, either ground-truth or predicted). SDA provides controlled gating of these sources and quantifies their standalone utility,

* Corresponding author.

E-mail address: inhi.kim@kaist.ac.kr (I. Kim).

<https://doi.org/10.1016/j.knosys.2025.115204>

Received 17 September 2025; Received in revised form 6 December 2025; Accepted 21 December 2025

Available online 23 December 2025

0950-7051/© 2025 Elsevier B.V. All rights are reserved, including those for text and data mining, AI training, and similar technologies.

marginal contributions, and interactions. This decomposition extends beyond ad hoc ablations and enables systematic and comparable attribution across datasets and model backbones.

This study is guided by the following research questions:

- **RQ1 (Past):** How does forecast accuracy evolve as the length of history increases, and where do diminishing returns occur?
- **RQ2 (Current):** How informative is the most recent observation in isolation compared to longer sequence-based inputs?
- **RQ3 (Prior):** How does the inclusion of destination information affect accuracy, and how does this vary with the quality of goal information?
- **RQ4 (Interactions):** To what extent are Past, Current, and Prior complementary or redundant under different prediction regimes?

This paper makes the following contributions:

1. We conduct a large-scale empirical study that decomposes and quantifies the contributions of Past, Current, and Prior across datasets and model classes.
2. We introduce the *Sequential Decomposition and Attribution* (SDA) protocol, a general-purpose evaluation framework that enables systematic source-level attribution through controlled gating and comparative metrics.
3. We derive actionable insights for trajectory forecasting, including when longer histories are beneficial, when current-only models suffice, and how priors can be effectively leveraged across horizons and contexts.

Our findings show that predictive gains from history often saturate with short sequences, that the most recent observation alone can be highly informative at short horizons, and that destination priors yield substantial improvements when available and accurate. The interactions among sources exhibit conditional complementarity, underscoring the importance of input-aware design for efficient and robust forecasting systems.

The remainder of this paper is organized as follows. [Section 2](#) reviews related work. [Section 3](#) introduces the SDA protocol. [Section 4](#) describes the experimental setup. [Section 5](#) presents attribution results. [Section 6](#) discusses implications, and [Section 7](#) concludes.

2. Related work

2.1. Sequence models for trajectory forecasting

Trajectory forecasting is widely formulated as sequence prediction over observed motion histories. Early deep learning approaches employed recurrent encoders, notably LSTM and GRU, to summarize an observation window and decode future states [3,9]. These models established strong baselines across pedestrian and vehicle domains, although plain recurrence may struggle to capture very long-term dependencies or complex multi-agent context.

To better represent spatiotemporal interactions, two prominent directions followed. Graph-based formulations represent agents as nodes with time-varying edges, enabling explicit reasoning over dynamic relations through message passing [4,10]. A related line of work integrates detection, tracking, and forecasting into unified pipelines [11–13], enabling joint reasoning over agent identities and future motion. In parallel, Transformer-based encoders apply self-attention to flexibly aggregate information within the observation window and across agents, targeting longer temporal horizons and heterogeneous dependencies [5, 6,8]. Both approaches enrich temporal and relational structure beyond purely recurrent encoders.

Generative and distribution-oriented formulations extend forecasting beyond deterministic extrapolation. Latent-variable models such as CVAE-based predictors generate multiple plausible futures by sampling from a learned latent space [14,15], while distribution-based methods

select or score candidate trajectories, or predict endpoint distributions before rollout [16,17]. These methods are particularly well suited to representing the inherent uncertainty and multimodality of human and vehicle motion, and can be combined with recurrent, graph, or Transformer backbones.

Empirical comparisons across benchmarks show that higher-capacity or longer-context encoders do not consistently outperform simpler recurrent or graph-based baselines; in some cases the latter remain competitive or even superior under comparable training conditions [7,8, 15,18]. Beyond supervised sequence prediction, several works couple forecasting with decision-making objectives or planning interfaces; however, adoption in standard trajectory benchmarks remains limited compared with supervised encoders.

2.2. Priors and contextual signals

In addition to temporal history, many approaches integrate external signals to guide or constrain forecasts. A central line of work introduces goal- or intent-conditioned priors, which bias predictions toward plausible destinations. Endpoint-conditioned models inject sampled or inferred goals into the decoder [7,19], while distribution-based approaches estimate candidate goals prior to rollout [16,17]. These priors are especially useful for long-horizon prediction, where extrapolation from short observation windows is often unreliable.

Contextual information offers complementary cues. Map-based encoders incorporate road topology, lane structure, and connectivity constraints to narrow the space of feasible trajectories [20,21]. Social-interaction modules capture group behaviors and collision avoidance, extending from pooling mechanisms [3] to graph-based reasoning [4, 10]. More recently, large-scale pretraining and language-modeling paradigms have been explored to provide semantic or intent-level priors for forecasting tasks, although their application is still in an exploratory stage. Collectively, these studies highlight the value of incorporating priors and context to reduce uncertainty and improve forecast realism.

2.3. Interpretability and attribution

Beyond improving accuracy, researchers have also sought to understand how forecasts depend on their inputs. Early analyses used ablation studies, removing components or signals and observing the effect on performance [3,9]. Attention-based encoders expose learned weights as saliency over tokens or agents, offering visual cues about influential observations [18,22]. However, attention redistributes importance among presented tokens, and several studies caution that attention weights may not correspond directly to explanatory importance or causal effect [23–25].

Gradient-based attribution adapts saliency and perturbation methods from vision to spatiotemporal data, attributing influence to frames or regions based on local sensitivity [26,27]. More formal approaches estimate marginal contributions with Shapley-style analyses or perturbation frameworks [28,29]. Each family offers different trade-offs: ablations are intuitive but vary in design, attention is easy to visualize but representation-level, gradient methods provide fine-grained sensitivity, and Shapley approaches are principled but computationally heavy. Together, they provide complementary perspectives on how temporal history, current state, and exogenous inputs affect forecasts.

2.4. Summary

Prior research on trajectory forecasting has progressed along three broad directions: the development of increasingly expressive sequence models from recurrent to graph- and attention-based encoders, the incorporation of priors and contextual signals such as goals, maps, and interactions, as well as jointly integration pipelines, and interpretability efforts ranging from ablations to attention, gradient-based, and Shapley-style analyses. Each line of work has expanded the field’s methodologi-

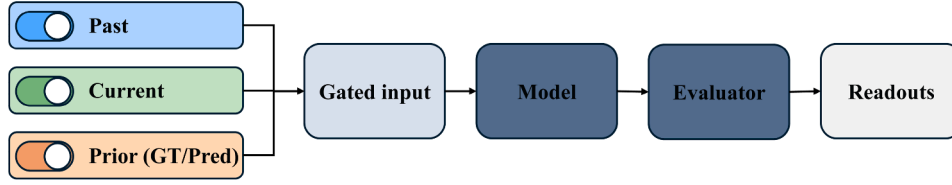


Fig. 1. Overview of the SDA protocol. The three sources—Past (P), Current (C), and Prior (R)—are selectively provided to the predictor through explicit availability gates. For each configuration, a separate model is trained with matched capacity and optimization settings, enabling attribution through controlled comparisons.

cal toolkit, yet several open questions remain. More powerful encoders have not consistently delivered superior accuracy, external signals have been integrated in diverse and often heuristic ways, and attribution studies have largely focused on saliency rather than systematically comparing the contributions of distinct input sources. These gaps motivate a closer examination of how past history, the current state, and destination priors each contribute to forecasting outcomes.

3. Methodology

We present the *Sequential Decomposition and Attribution* (SDA) protocol to analyze how different temporal input sources contribute to sequence forecasting. SDA operates at the input level by explicitly controlling the availability of each source before encoding or fusion. An overview is provided in Fig. 1.

3.1. Decomposition of temporal sources

The forecasting model receives three distinct input sources at prediction time:

$$P = \{x_{t-L}, \dots, x_{t-1}\}, \quad C = x_t.$$

Here, *Past* refers to a fixed-length window of L historical observations, and *Current* denotes the most recent state.

The third source, *Prior*, represents goal-related information in the same spatial frame as the forecast. When available, it can be the ground-truth destination r^* , or alternatively a predicted goal $\hat{r}$ obtained from a goal estimator:

$$R \in \mathcal{R}, \quad R = r^* \quad \text{or} \quad R = \hat{r} = g_\phi(P, C, \text{ctx}),$$

where g_ϕ leverages past trajectories, current state, and optional context (e.g., map or semantics). Its prediction error

$$e = \hat{r} - r^*, \quad \mathbb{E}[e] = b, \quad \text{Cov}(e) = \Sigma$$

captures bias and uncertainty in prior estimation. Within SDA, R is an observed conditioning input—either ground-truth or model-predicted—and is distinct from a Bayesian parameter prior or a Markov transition assumption; our attribution targets empirical effects of availability and quality, irrespective of the inference paradigm. For clarity, throughout this paper the term “Prior” refers solely to destination-related conditioning inputs used by forecasting models; it does not denote a Bayesian prior distribution. Our use of the term concerns the empirical effect of providing or withholding goal information, independent of any probabilistic-inference interpretation.

3.2. Gated input protocol

To systematically study source contributions, we control input availability via hard binary gates applied at the input level. When a gate is enabled, the source is fed into the model; when disabled, it is replaced with a neutral placeholder that blocks both feature propagation and gradient flow.

Concretely, P supplies $\{x_{t-L}, \dots, x_{t-1}\}$; when gated off, it becomes an empty sequence of length zero (no tokens). C supplies x_t ; when gated off, it is replaced by a zero vector of matching dimension. R supplies a

goal vector in the same normalized coordinate system as the forecast; when gated off, it is replaced by a zero goal vector.

Implementation-wise, disabling a source does not introduce padded tokens or variable-length padding into the encoder. When P is disabled, we set $L=0$ so that no history tokens are passed to the LSTM/Transformer encoder, and the past branch is replaced by a fixed zero embedding. When C or R are disabled, their encoders receive a constant zero input shared across all trajectories, ensuring that no learnable “special token” is introduced and that zero vectors do not carry unintended information. All configurations therefore preserve the same positional encoding and sequence structure; differences in performance across SDA conditions reflect only the availability of actual information sources rather than artifacts from padding, masking, or attention behavior.

Each subset $S \subseteq \{P, C, R\}$ defines a unique source configuration. For every such S , we train a separate model using the same backbone family, parameter budget, optimizer, and data splits so that observed performance differences are attributable to input availability. Although many forecasting models adopt attention or other soft weighting to modulate importance, these approaches only redistribute emphasis among tokens that are present and cannot enforce the absence of an entire source. Our hard gating enforces strict exclusion; any internal attention then operates only over the inputs present in a given configuration.

3.3. Evaluation and attribution

Let f_θ be a trained forecasting model and $\mathcal{M}$ an evaluation metric such as ADE or FDE. Denote the full source set by $N = \{P, C, R\}$. For any subset $S \subseteq N$, we define the expected error as

$$M(S) = \mathbb{E}[\mathcal{M}(f_\theta(\tilde{x}(S)), \mathbf{y})],$$

where $\tilde{x}(S)$ indicates input with sources in S retained and others neutralized.

We consider three forms of attribution. First, the standalone utility of a source i is given by its performance in isolation, denoted $M(\{i\})$. Second, the marginal contribution of a source is defined relative to a base set B :

$$\Delta_{i|B} = M(B) - M(B \cup \{i\}), \quad i \notin B,$$

so that a positive value indicates a reduction in error when adding i . Finally, the interaction between two sources is measured by comparing their joint performance with the better of their single-source baselines:

$$I_{i,j} = \min\{M(\{i\}), M(\{j\})\} - M(\{i, j\}),$$

where a negative value suggests synergy, and a positive value implies redundancy.

Conditional interactions can also be defined. For example,

$$I_{C,R|P} = \min\{M(\{P, C\}), M(\{P, R\})\} - M(\{P, C, R\}),$$

measures how C and R interact when P is held fixed.

A general form for arbitrary subset interactions is provided in Appendix C, where conditional pairwise metrics emerge as special cases under fixed context.

4. Experimental design

We instantiate SDA through controlled experiments on representative datasets and model classes. This section specifies the forecasting

Table 1
Dataset statistics and preprocessing.

Dataset	#Traj	Sampling (s)	Avg. length	Type
Porto	257,964	15	29	Taxi
ETH	337	0.4	25	Pedestrian

task, datasets, model backbones, and baselines, as well as the evaluation and training protocol. The design aims to attribute performance to information sources rather than confounders.

4.1. Task and factorial conditions

The task is *trajectory forecasting*: given an observed prefix, the model predicts a future horizon of length H . Following the SDA decomposition, we selectively enable three information sources: the *Past* (P) as a history of length L , the *Current* (C) as the last observed state, and the *Prior* (R) as destination information when available. Under this convention, a purely current-based setting is defined as $L=0$.

To study the effect of prior quality in a controlled manner, we simulate predicted priors by perturbing the ground-truth destination with zero-mean isotropic Gaussian noise,

$$\tilde{r} = r + \epsilon, \quad \epsilon \sim \mathcal{N}(0, \rho^2 I),$$

and vary ρ from 0 (oracle) upward. This mechanism provides a distributionally simple, single-factor degradation of prior quality, enabling clean attribution without confounding architectural or contextual differences. Real goal predictors may exhibit structured, map-dependent, or multimodal errors; incorporating such estimators into SDA is possible but would introduce model-specific biases that obscure the source-level effects studied here.

Unless otherwise noted, we use $H \in \{1, 3, 5, 7, 9, 11\}$. For history length, Porto uses $L \in \{1-14, 20, 50, 100\}$ and ETH uses $L \in \{1-9\}$. Specific subsets are reported alongside the results in Section 5.

The SDA decomposition is applied to ego-only inputs: C refers solely to the ego agent's last observed state and does not encode neighbor trajectories or social-interaction cues. Although multi-agent interactions are present in datasets such as ETH, incorporating them constitutes a modeling choice beyond the scope of source-level attribution. SDA can naturally accommodate such settings by treating neighbors' history or instantaneous state as additional information channels; exploring these extensions is left for future work.

4.2. Datasets

Porto (vehicle GPS). Taxi trajectories in an urban road network. Preprocessing includes projection to UTM coordinates, resampling at 15s intervals, and filtering outliers and stationary segments. Chronological split: 6/2/2 (train/val/test).

ETH (pedestrian). Pedestrian trajectories in crowd scenes. Pixel coordinates are mapped to meters via homographies and resampled at 0.4s. Splits follow a 6/2/2 (train/val/test) reorganization by scene.

Table 1 reports dataset statistics and preprocessing details.

4.3. Models and baselines

LSTM. A 1–2 layer recurrent encoder. **Transformer.** An encoder-only model with causal self-attention. Both backbones follow standard formulations and their configurations are listed in Table 2. Because attribution is analyzed within each backbone rather than across backbones, we set the LSTM and Transformer architectures to comparable parameter scales to avoid large capacity disparities that would otherwise confound the interpretation of information-source effects.

Reference baselines: **Last-Point** (copy last observed state), **Linear Extrapolation** (velocity projection), **Current-MLP** (small MLP on the last state).

Table 2
Model configurations (example values).

Model	Params	Hidden	Heads	Dropout
LSTM	~0.5M	128	–	0.1
Transformer	~0.5M	128	4	0.1
Current-MLP	~0.05M	64	–	0.1

Table 3
Training hyperparameters.

Batch size	LR	Epochs	Patience	Seed
64	1e-4	200	10	42

4.4. Evaluation and training

Forecasting accuracy is measured by two displacement-based metrics: **ADE** (Average Displacement Error) and **FDE** (Final Displacement Error). Training hyperparameters are provided in Table 3. All models are trained with Adam (learning rate 1e-4, batch size 64) for up to 200 epochs, with early stopping on validation ADE (patience 10). All experiments use a fixed random seed (42). Unless otherwise stated, results are single-run point estimates ($K=1$).

5. Experimental results

This section reports empirical results following the protocols in Section 4 and the attribution formalism in Section 3. Unless otherwise stated, all experiments follow the single-seed protocol ($K=1$) described in Section 4. To assess stability, all configurations in the overall-performance table (Table 4) are additionally evaluated under three random seeds, and the aggregated mean $\pm$ std results are reported in Appendix A. The observed variability is modest across both datasets, supporting the single-seed setting for the remaining analyses (history-length sweeps, prior-quality curves, and interaction studies). We also report a current-only regime defined by $L=0$ (cf. Section 4), which does not conflict with the candidate L sets used elsewhere.

5.1. Overall performance

We evaluate the short-horizon setting ($H=5$; metrics in meters) using observation windows $L=20$ for **Porto** and $L=9$ for **ETH**, while the current-only regime uses $L=0$. Rows involving destination information adopt ground-truth goals and are reported as non-operational upper bounds ($\dagger$). Unless otherwise stated, aggregates in this subsection use the *LSTM* backbone. We omit a Transformer aggregate table because subsequent length sweeps and interaction diagnostics (Section 5.2 and Section 5.4; see also Section 6) confirm that cross-source *relative* trends are consistent across backbones, so a second table would be redundant.

Table 4 consolidates simple current-only baselines with SDA-gated inputs for side-by-side comparison. On **Porto**, the lightweight *Current-MLP* (119.836 ADE) marginally improves upon *Last-Point* (122.275 ADE) and clearly outperforms *Linear* (135.250 ADE), suggesting that a small-capacity model exploiting the last state can already mitigate road-network noise and stop-and-go behaviors more effectively than naive copying or constant-velocity extrapolation. On **ETH**, the *Linear* baseline (0.302 ADE; 0.508 FDE) is unexpectedly competitive at $H=5$, lying within about 2% of the LSTM current-only ADE (0.296). This reflects near-inertial short-horizon pedestrian motion; despite this proximity in ADE, *Linear* remains inferior in FDE and is surpassed once temporal context or goal priors are fused. Note that our *Current-MLP* is intentionally small (~0.05M parameters) and consumes only the last state; in ETH's short-horizon regime, this bias favors linear projection.

Turning to SDA-gated inputs, the current state alone is broadly competitive across both datasets, while fusing short history with the current state yields consistent gains. On **Porto**, current-only improves over

Table 4

Overall performance at short horizon ($H=5$). Entries show ADE $\downarrow$ with FDE in parentheses. Porto/ETH use $L=20/9$; *current-only* uses $L=0$.

Method (inputs)	ADE $\downarrow$ (FDE)	
	Porto	ETH
Last-Point (C)	122.275 (194.464)	1.733 (2.873)
Linear Extrapolation (C)	135.250 (231.608)	0.302 (0.508)
Current-MLP (C)	119.836 (190.168)	1.645 (2.779)
Past-only (P)	103.424 (148.920)	0.293 (0.420)
Current-only (C)	95.442 (152.708)	0.296 (0.437)
Past + Current (P+C)	82.179 (129.763)	0.239 (0.355)
Current + Prior (C+R [†])	86.175 (130.677)	0.275 (0.400)
Past + Prior (P+R [†])	87.513 (118.572)	0.270 (0.376)
Past + Current + Prior (P+C+R [†])	<u>72.765</u> (108.623)	<u>0.222</u> (0.322)

Rows marked [†] use ground-truth goals and serve as non-operational upper bounds. Note. ETH baseline RMSE (omitted above): Last-Point 1.940, Linear 0.423, Current-MLP 2.229.

past-only (95.442 vs. 103.424 ADE), and Past + Current reaches 82.179 ADE; adding oracle goals further reduces error to 72.765 ADE, indicating substantial headroom when reliable destination priors are available. Relative to the simple baselines, the LSTM current-only improves ADE by about 20.4% over *Current-MLP* and about 29.4% over *Linear*; adding short history (P+C) yields a further about 31.4% gain versus *Current-MLP*. On *ETH*, past-only and current-only are very close (0.293 vs. 0.296 ADE), consistent with stronger stochasticity and scene-level constraints; nevertheless, P+C achieves 0.239 ADE, and oracle goals reduce it to 0.222 ADE. Compared with *Linear*, P+C improves ADE by about 20.9%, and the oracle prior provides an additional margin. These patterns—strong last-state signal, early gains from short histories, and substantial improvement from accurate goal information—are examined in detail by the length sweeps (Section 5.2) and prior-quality analyses (Section 5.4). Finally, although this subsection summarizes LSTM aggregates, later sections verify that Transformers exhibit the same ordering across {P, C, R} regimes.

5.2. Past contribution: History-length curves

To isolate the role of historical observations, we vary the history length L while keeping the prediction horizon $H \in \{1, 3, 5, 7, 9, 11\}$ fixed per curve and using the same training protocol as in Section 5.1. Fig. 2 reports results on *Porto*; the analogous plots for *ETH* appear in Appendix B. Each row corresponds to one backbone (LSTM, Transformer). The left panels plot absolute ADE versus L , and the right panels show the *relative* improvement over $L=0$ (current-only). The shaded region marks the saturation zone (defined as achieving at least $p=90\%$ of the total attainable improvement), and the vertical dashed line indicates the corresponding L^* .

On *Porto*, the curves consistently exhibit diminishing returns. Adding a few historical steps on top of the current state yields substantial accuracy gains, but enlarging L beyond that range brings only marginal improvements, and the relative-improvement curves flatten quickly. This pattern holds across both short and long horizons: longer horizons benefit slightly more from additional history, yet saturation occurs early (typically $L^* \leq 8$). *ETH* shows the same tendency, albeit less smoothly because of the smaller dataset and higher stochasticity of pedestrian trajectories; most of the benefit is again realized within the first few history frames Appendix B.

From a deployment perspective, increasing L from 0 to a small range (for instance, $L \in [1, 3]$) typically provides a large improvement at limited additional cost, whereas further increases deliver little additional accuracy while computation continues to rise. In practice, for a given horizon H and backbone, selecting the smallest L in the early plateau that keeps validation ADE within about 1–2% of the best value offers a reliable compromise. A formal complexity proxy and the corresponding trade-off plots are presented in Section 6 (Fig. 9).

Table 5

Prior degradation summary on *Porto* with $L=20$. Δ ADE is the oracle gain over the no-prior baseline. ρ^* is the break-even noise level where the degraded prior equals the baseline. ρ_{50} is the half-gain point; parentheses show the expected absolute prior error in meters.

Backbone	H	Baseline	Oracle	Δ ADE	ρ^* / ρ_{50} (m)
LSTM	5	84.5	76.1	8.4	1.76 / 0.52 (~373)
Transformer	5	89.7	77.7	12.0	1.84 / 0.50 (~359)
LSTM	11	139.5	119.4	20.1	9.23 / 0.77 (~553)
Transformer	11	140.6	121.2	19.4	2.69 / 0.59 (~424)

5.3. Current contribution: Strong last-state signal

To quantify the benefit of the most recent observation, we fix the history length to $L=5$ and, for each prediction horizon $H \in \{1, 3, 5, 7, 9, 11\}$, compare three input settings on *Porto*: *past-only* (Past), *current-only* (Current), and the fusion *Current + Past* (Current + Past). The top row of Fig. 3 plots ADE versus H for LSTM and Transformer. Across all horizons, **Current + Past** (green) consistently outperforms both **Past** (red) and **Current** (blue), confirming that the last state and a short history carry complementary information.

The bottom row shows the marginal gains of adding **Current** on top of **Past**:

$$\Delta_{\text{abs}}(H) = \text{ADE}_{\text{Past}}(H) - \text{ADE}_{\text{Current} + \text{Past}}(H),$$

$$\Delta_{\text{rel}}(H) = \frac{\Delta_{\text{abs}}(H)}{\text{ADE}_{\text{Past}}(H)} \times 100\%.$$

Two observations are consistent across backbones. (i) **Large short-horizon gains.** At $H=1$ the absolute reduction is about 31 m ($\approx 47\%$ for LSTM; $\approx 45\%$ for Transformer), showing that the last state provides strong instantaneous heading and speed cues. (ii) **Steady diminishing returns with longer horizons.** As H grows, both absolute and relative gains decrease smoothly, reaching about 19–23 m and 12–14% by $H=11$. Minor fluctuations at large H are small compared with the overall monotonic trend.

Takeaway. For practical deployments, including the **Current** signal yields the biggest payoff at short horizons; for longer horizons the marginal value persists but is modest, while the fusion **Current + Past** remains a robust default.

5.4. Prior contribution: Ground-Truth vs. predicted goals

We study how destination priors influence accuracy by degrading the prior with zero-mean isotropic noise of level ρ in the normalized (x, y) space (details in Section 4). Larger ρ means a noisier prior. Fig. 4 reports ADE as a function of ρ for LSTM and Transformer at two horizons $H \in \{5, 11\}$ while fixing the observation window to $L=20$. Each panel also shows the no-prior baseline (dashed), the oracle prior at $\rho=0$ (star), and two operating points: (i) the break-even point ρ^* where the degraded prior matches the baseline, and (ii) the half-gain point ρ_{50} where one half of the oracle improvement remains. The light band highlights the practically relevant low-noise range $\rho \in [0, 2]$.

To summarize the curves quantitatively, Table 5 lists the baseline ADE, the oracle ADE, the oracle gain Δ ADE, the break-even ρ^* , and the half-gain ρ_{50} with its approximate value in meters. We convert ρ to meters using the model’s normalization scale via $\mathbb{E}[\|e\|_2] \approx 1.253 \bar{\sigma} \rho$; on *Porto* we estimate $\bar{\sigma} \approx 573$ m, hence $1 \rho \approx 718$ m.

5.5. Disentangling complementarity: Past, current, and prior

We next examine how the three sources combine in practice, using the utility, marginal, and interaction metrics defined in Section 3. Rather than reintroducing notation, we focus on the qualitative behaviors that account for the aggregate trends reported earlier.

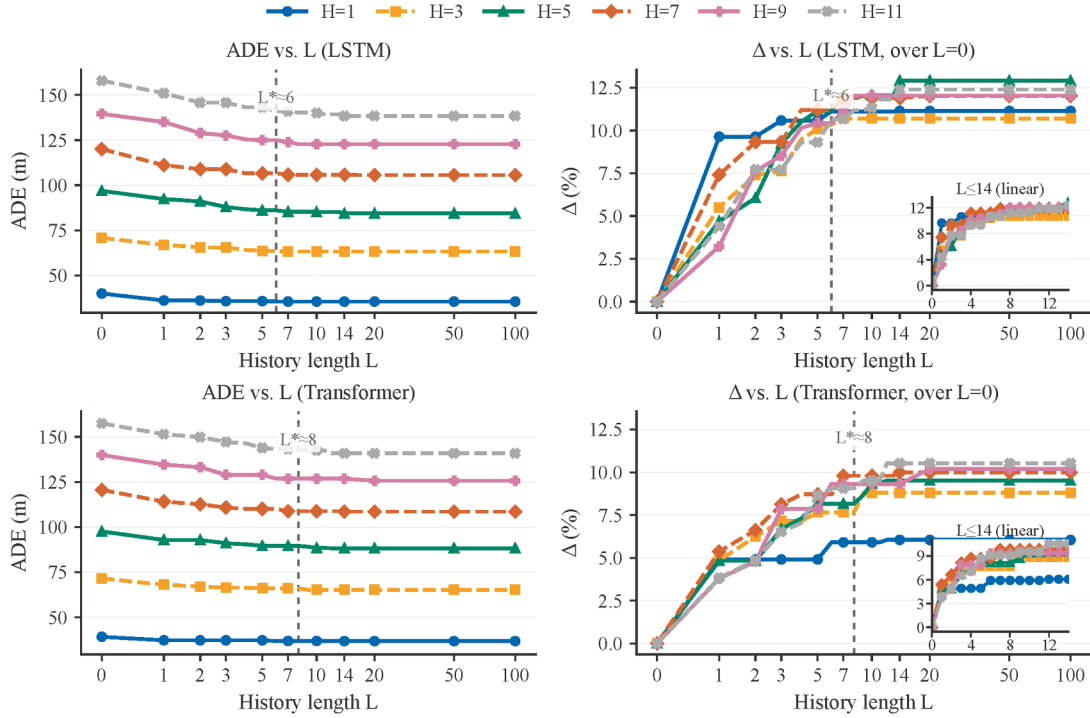


Fig. 2. Porto - History-length sweep. Left: ADE versus L for LSTM and Transformer at multiple horizons H . Right: relative improvement over $L=0$ (current-only). The shaded band indicates the saturation zone where coverage reaches $p=90\%$ of the total attainable improvement, and the vertical dashed line marks the corresponding L^* . Curves show early gains followed by clear saturation across horizons and backbones.

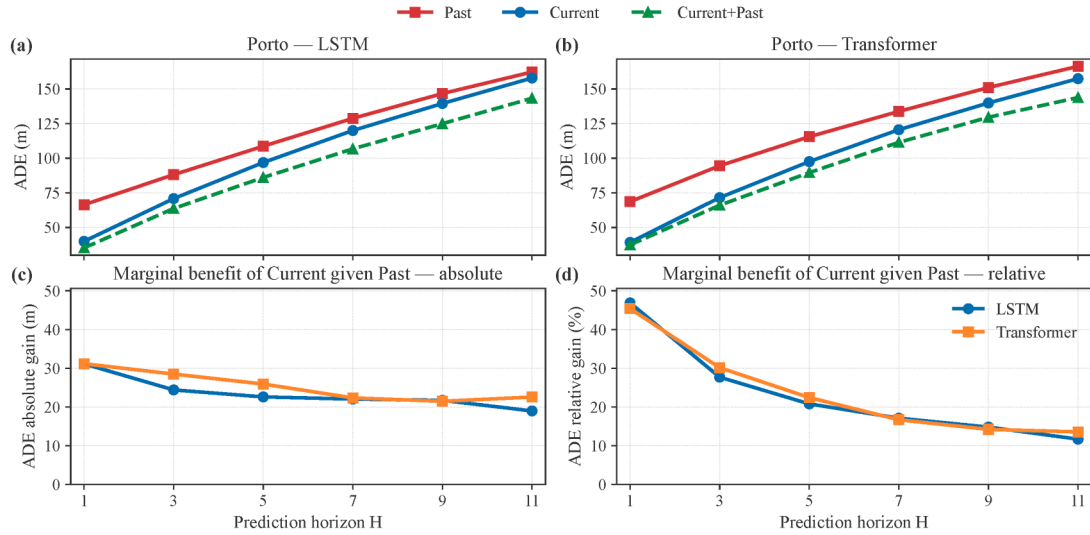


Fig. 3. Porto, fixed $L=5$. Top: ADE vs. horizon H for Past, Current, and their fusion Current + Past under LSTM and Transformer backbones. Bottom: Marginal benefit of adding Current on top of Past (absolute and relative). The last state is highly informative for small H ; gains taper off as H increases, while Current + Past consistently dominates either single-source input.

A first comparison contrasts settings with and without priors across a range of history lengths. With a reliable destination cue, ADE decreases uniformly and the saturation point in L shifts left, meaning that comparable accuracy can be achieved with shorter histories once R is present. This shift clarifies how priors effectively substitute for a portion of the temporal context. The full curves for LSTM and Transformer are shown in Fig. 5.

A second view focuses on the joint use of P and C . Across horizons, adding the current state on top of history leads to sizable reductions

at short horizons and non-negligible gains even when L is large. This indicates that instantaneous heading and speed cues in C are not fully recoverable from P alone, and that encoders benefit from treating the last state with special care rather than absorbing it into a long sequence. The complementarity between P and C also tends to increase with L . These patterns are visualized in Fig. 6.

Conditioned on the current state, priors play a different role. When histories are short, adding R yields the largest improvements, and the conditional complementarity diminishes as L grows. This behavior

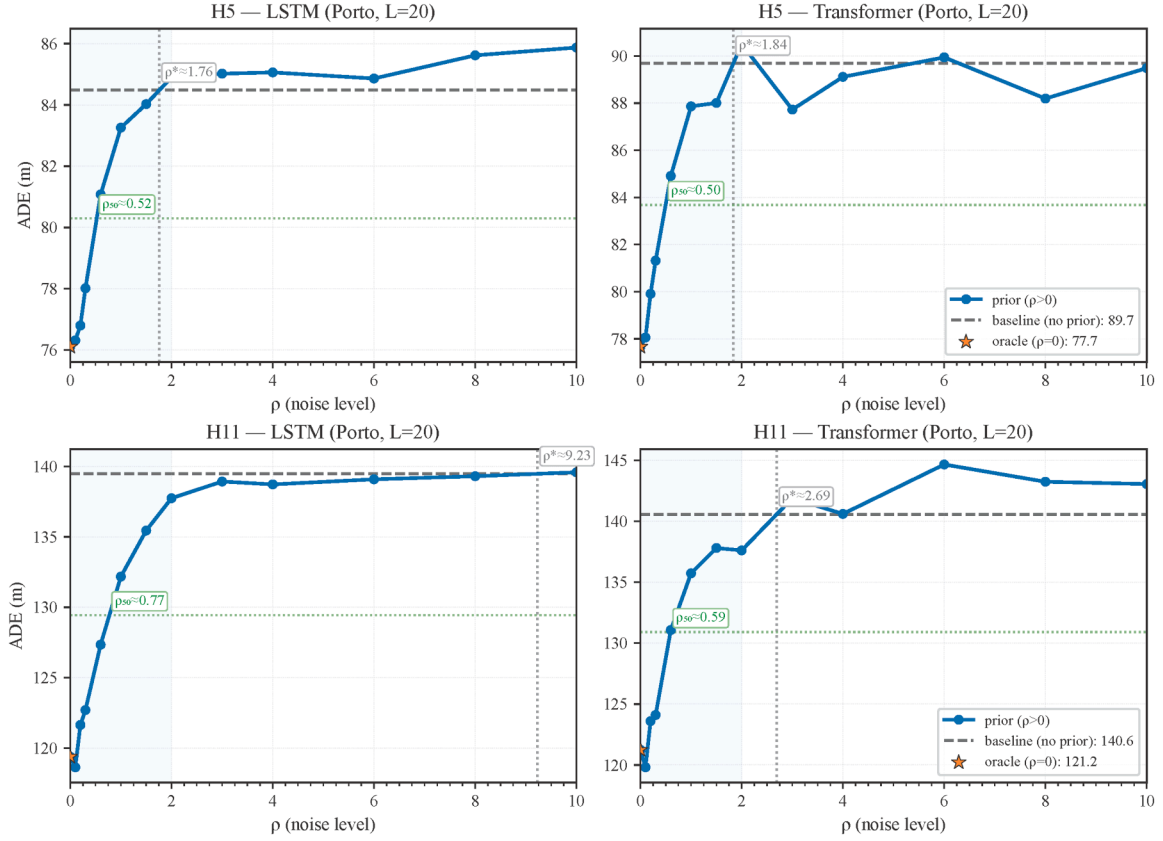


Fig. 4. Porto, $L=20$. ADE versus prior-noise level ρ for LSTM and Transformer at $H \in \{5, 11\}$. Dashed: no-prior baseline. Star at $\rho=0$: oracle prior. Vertical dotted lines mark ρ^* (break-even). Horizontal dotted lines mark the half-gain level that defines ρ_{50} . The shaded region emphasizes the low-noise regime $\rho \in [0, 2]$.

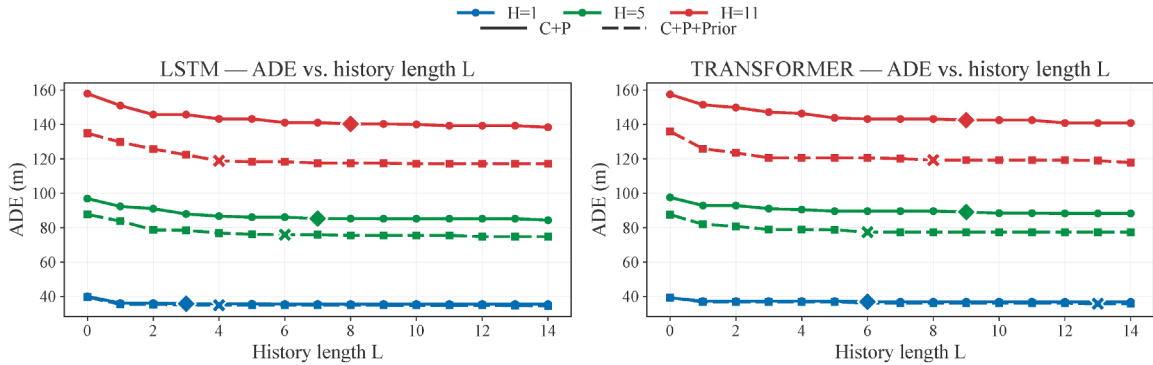


Fig. 5. Porto. ADE vs. history length L with prior (C+P+R, dashed) and without prior (C+P, solid) for LSTM and Transformer at $H \in \{1, 5, 11\}$. Markers indicate L^* under a $p=90\%$ coverage rule. Priors decrease ADE and reduce the history needed to reach saturation.

matches the intuition that goal information is most valuable when the temporal context is limited or when the horizon is long. Fig. 7 summarizes these tendencies.

Finally, conditioned on history, the interplay between C and R becomes increasingly important as the horizon grows. With L fixed (e.g., $L=5$), the gain from introducing R is strongest when the destination lies near the current position and decreases with distance, while the aggregate complementarity between C and R rises with H . Fig. 8 visualizes these effects.

Taken together, these observations suggest a horizon- and context-aware division of labor: C stabilizes short-term kinematics, R provides long-range guidance, and P fills in residual dynamics, with diminishing returns once L passes the early plateau.

6. Discussion

Building on the experimental results in Section 5, this section provides analysis and interpretation of the findings. We discuss the implications of diminishing historical returns, the dominant role of current and goal-related signals, the structured complementarity among sources, and broader implications, limitations, and future directions.

6.1. Historical redundancy and information saturation

The history-length sweeps indicate that predictive gains saturate quickly once the current state is included (Section 5.2). On Porto, most of the attainable improvement is realized by small L before the curves

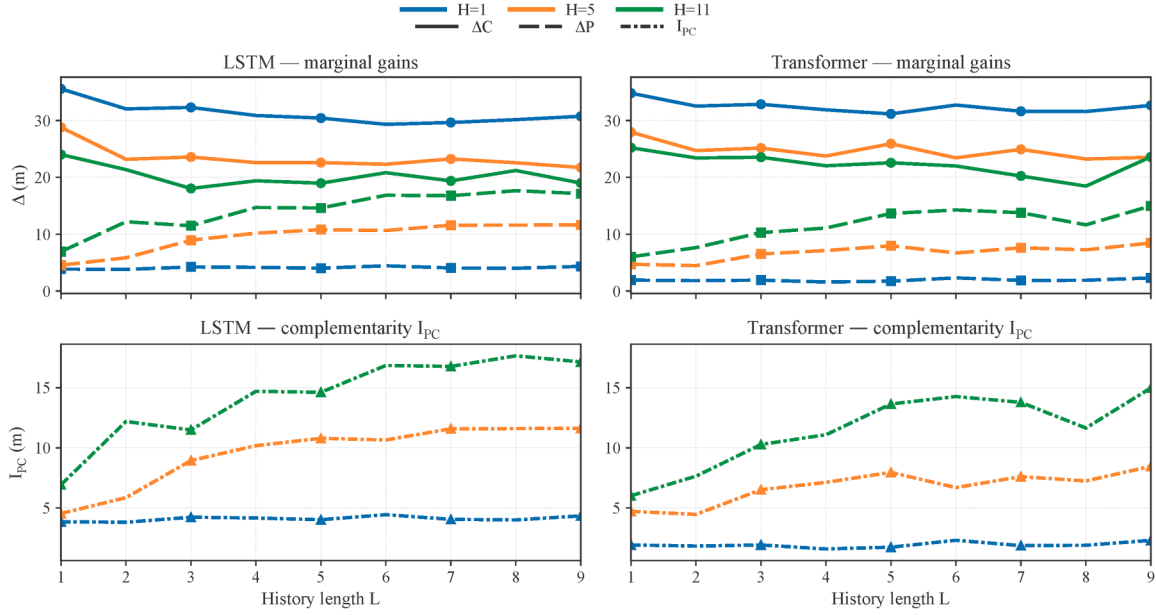


Fig. 6. P-C interaction (Porto). Top: marginal gains of adding *Current* on top of *Past* and vice versa across L and horizons. Bottom: complementarity $I_{P,C}$ increasing with L , indicating that longer histories enable better use of the last state.

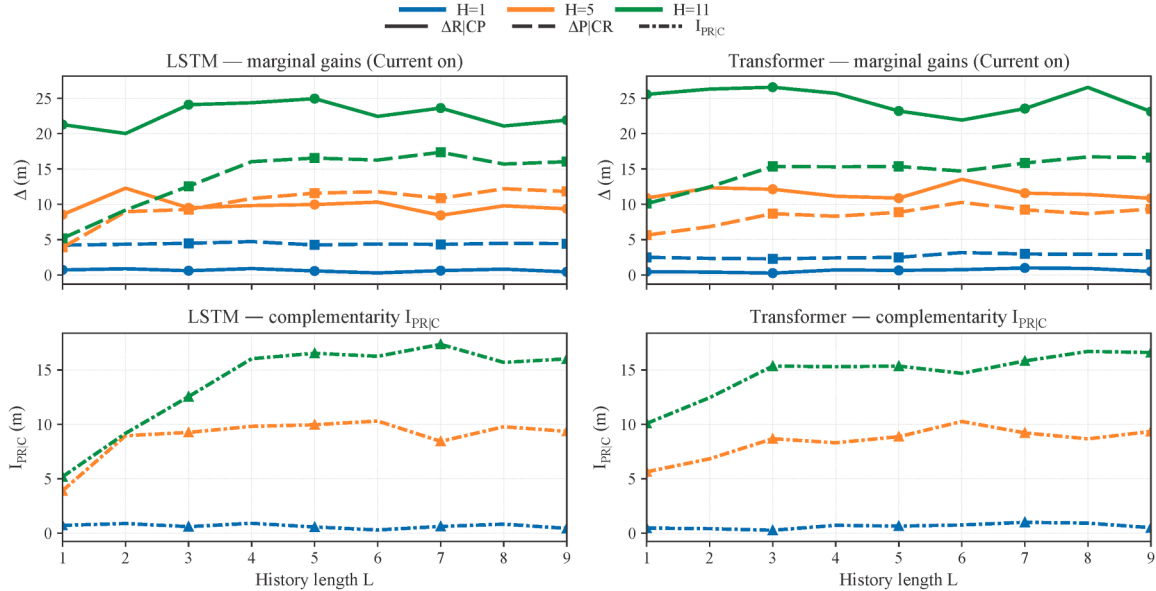


Fig. 7. R-P interaction conditioned on C (Porto). Top: marginal gain of *Prior* on $P \cup C$ (solid) and of *Past* on $C \cup R$ (dashed) across L and horizons. Bottom: conditional complementarity $I_{R,P|C}$ largest at small L , highlighting the value of priors when the temporal context is limited.

plateau; ETH exhibits the same tendency despite higher stochasticity. These observations are consistent with recent evidence that enlarging temporal context does not guarantee better sequential modeling performance [30].

We use an analytic proxy for computational cost (not wall-clock), parameterized by L . With hidden width and batch size fixed, the stepwise computation of an LSTM encoder grows approximately linearly with L , whereas the dominant self-attention term of a Transformer grows quadratically:

$$C_{\text{LSTM}}(L) = aL + b, \quad C_{\text{Trans}}(L) = \alpha L^2 + \beta L + \gamma. \quad (1)$$

Fig. 9 plots ADE against a *normalized* version of this cost proxy (so that $L=0$ maps to 1) for Porto and ETH. The Pareto frontiers show that increasing L from 0 to a small range (e.g., $L \in [1, 3]$) produces a large accuracy gain at limited additional cost; beyond that range, ADE im-

provements quickly saturate while the proxy cost continues to rise. In practice, for a given horizon H and backbone, choosing the smallest L in the early plateau that keeps validation ADE within about 1-2% of the best value offers a reliable accuracy-efficiency compromise. For larger horizons or more complex dynamics, the frontier shifts to the right, yet the same marginal pattern is observed; on small, noisy datasets such as ETH, the curves are less smooth, but most benefits still come from the first few frames.

Implication. Prefer truncating or compressing history (e.g., adaptive token selection, temporal dropout, learned masking).

6.2. Dominant role of current and goal information

The results in Section 5.3 and Section 5.4 together suggest a clear division of labor between the most recent observation and destination

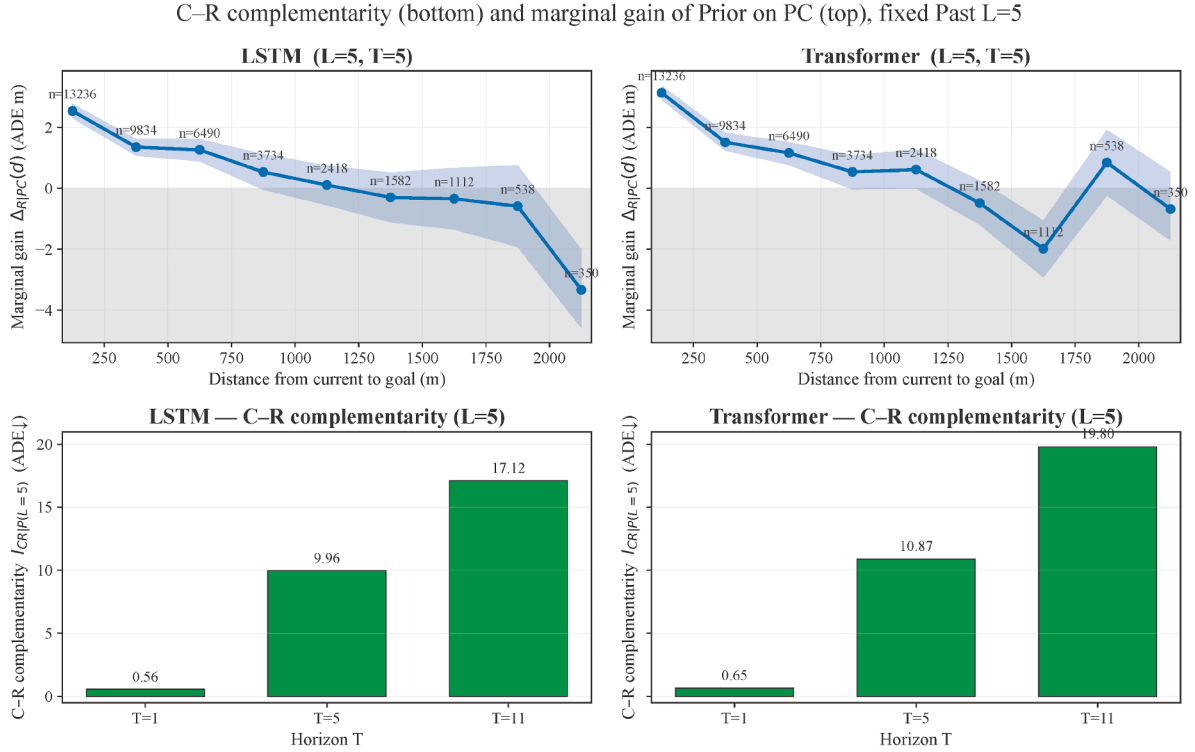


Fig. 8. C-R interaction conditioned on P (Porto, $L=5$). Top: marginal gain of R on $P \cup C$ as a function of distance-to-goal; benefits are strongest near the goal. Bottom: complementarity $I_{C|P}$ increases with the horizon H , indicating strong long-horizon synergy.

priors. For short horizons, the current state carries highly predictive heading and speed cues: adding C on top of P yields the largest absolute and relative gains at $H=1$ and remains beneficial, though with diminishing returns, as the horizon increases (Fig. 3). For longer horizons, goal information becomes decisive. The degradation study in Fig. 4 shows that oracle priors produce substantial improvements, and these benefits decay smoothly with noise. The summary in Table 5 quantifies the break-even level ρ^* and the half-gain point ρ_{50} across backbones and horizons.

A complementary view is obtained by comparing ADE- L curves with and without priors. With reliable R, the entire curve shifts downward and the saturation point in L moves left, indicating that similar accuracy can be achieved with shorter histories once goal information is available. This effect is consistent across backbones and horizons and is visualized in Fig. 5, which aggregates LSTM and Transformer results at $H \in \{1, 5, 11\}$. In practical terms, priors reduce the reliance on long observation windows, while the current state stabilizes short-term kinematics; the two signals therefore act in concert when both near-term control and long-horizon intent matter.

These observations motivate input-aware designs. For short-term forecasting and reactive control, encoders should preserve C with high fidelity and fuse it early. For planning horizons and strategic maneuvering, improving the quality of R (e.g., via better goal predictors or map/semantic cues) directly translates into downstream accuracy, and even moderately noisy priors remain useful up to ρ_{50} (Table 5). When both requirements coexist, models should allocate capacity to a strong current pathway and a robust, uncertainty-aware prior pathway, while compressing or selecting history once the early plateau in L is reached (cf. Fig. 5).

6.3. Complementarity versus redundancy

The interaction patterns above admit a coherent interpretation. The current state augments history in a way that remains visible even when

several past steps are available, which argues for encoding C with high fidelity and fusing it early rather than burying it in a long sequence. Priors reduce the effective need for long histories. When destination cues are accurate, the same accuracy can often be reached with much shorter observation windows, and the saturation point in L consistently moves left. At the same time, the synergy between current and prior information strengthens as the prediction horizon increases, reflecting a natural division of roles between near-term kinematics and long-horizon intent.

For design, the implication is straightforward. History should be compressed or selected aggressively once the early plateau is reached; the current state should be modeled with explicit, high-weight pathways; and priors should be weighted more heavily when horizons are long or uncertainty is high. Fig. 10 distills these relationships into a single schematic that annotates how complementarity changes with history length and horizon.

6.4. Distinction from standard ablations

Standard ablation studies perturb a *single* trained model by masking or removing selected inputs or components, so any observed performance change reflects how that particular shared-parameter architecture compensates for the missing signal through correlated features, residual temporal structure, or representation sharing. SDA differs in its experimental stance: instead of perturbing one model, it defines distinct information-availability conditions and trains an *independent* model for each subset $S \subseteq \{P, C, R\}$ under matched backbone family, comparable parameter scale, optimization settings, and data splits. Hard input gating ensures that a source is either fully present or entirely absent with no cross-source parameter sharing. Consequently, differences in performance arise from the intrinsic predictive value of the information sources themselves, rather than from perturbation dynamics within a single architecture.

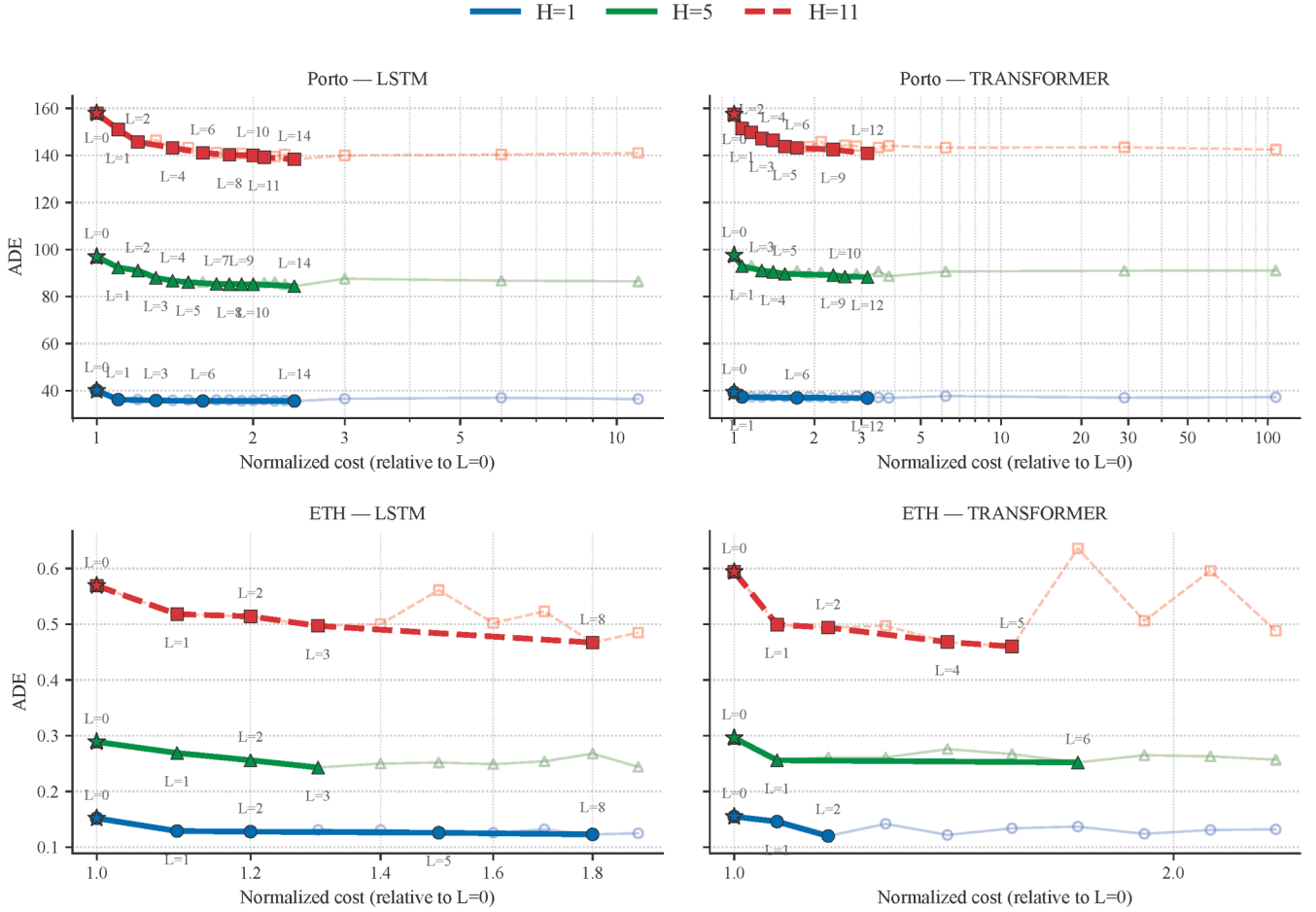


Fig. 9. Accuracy-efficiency trade-off on **Porto** and **ETH** using an analytical complexity proxy that depends on L . Each panel shows ADE versus a cost proxy normalized such that $L=0$ has value 1 (approximately linear in L for LSTM and quadratic for Transformer). Colors denote horizons $H \in \{1, 5, 11\}$, and markers are annotated with the corresponding L . Most of the accuracy gain is achieved at small L ; subsequent increases bring limited benefits while the proxy cost grows rapidly.

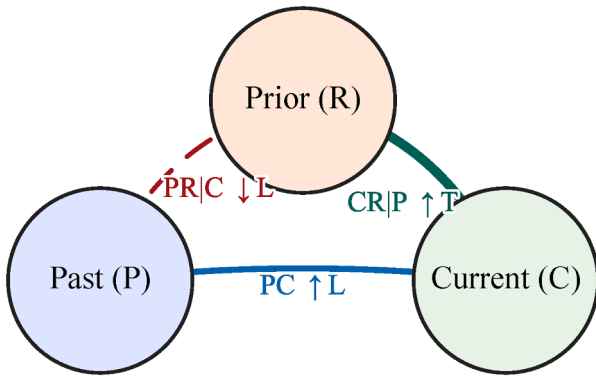


Fig. 10. Summary of input complementarity. Schematic view of Past (P), Current (C), and Prior (R) interactions. Thick edges indicate strong synergy; dashed edges denote conditional or weak complementarity. Labels annotate how complementarity changes with history length L and horizon H .

6.5. Broader implications

For model design. Allocate capacity adaptively: (i) compress or select history tokens; (ii) use current-aware encoders to capture instantaneous dynamics; (iii) integrate prior-aware planners for goal-conditioned long-horizon prediction. Horizon-aware gating or curriculum schedules can modulate reliance on P, C, and R. Importantly, while

our experiments adopt fixed, manually specified on/off gates to isolate information sources, the SDA protocol is agnostic to the particular gating or token-selection mechanism: learned soft gates, adaptive horizon-aware gating, or differentiable token-selection modules can be substituted without altering the attribution definition. In addition, SDA suggests maintaining explicit and separately parameterized pathways for current- and prior-related signals, enabling architectures to allocate capacity according to their distinct temporal roles.

For data collection. Extending observation windows is less impactful than curating reliable priors and contextual signals (e.g., destinations, maps, semantics). Goal/intent annotations often yield larger returns than additional raw history.

For deployment. Shorter histories reduce latency and memory, facilitating edge deployment. Current-aware modules provide strong short-horizon baselines for safety-critical control, whereas prior-aware modules support strategic planning. Privacy is improved when storing compressed summaries instead of long raw trajectories.

6.6. Limitations and future directions

The study covers two datasets (Porto, ETH) and two backbones (LSTM, Transformer); broader domains, including highways, multi-modal mobility, maritime and aerial navigation, as well as non-mobility sequential prediction settings, are needed to establish the external validity of the observed information contributions beyond the two urban

datasets studied here. Accuracy (ADE/FDE) is our primary focus because these displacement metrics are standard, but they primarily capture geometric error; future work should incorporate robustness, comfort or smoothness, collision risk, and task-dependent utility to obtain a more holistic evaluation. Priors are simulated via isotropic Gaussian noise, which does not capture the structured and map-dependent errors of real goal predictors; learned estimators may introduce non-Gaussian, systematic biases that interact differently with P and C.

Promising directions include: (i) adaptive prior weighting driven by horizon and uncertainty; (ii) structural regularizers encoding complementarity (e.g., information-theoretic or sparsity constraints across sources); (iii) multi-agent extensions where priors encode socially-aware intent and interaction, allowing SDA to quantify how agent-to-agent coupling modifies the contributions of P, C, and R; (iv) edge-oriented designs coupling history compression with calibrated goal inference; and (v) methodological extensions connecting SDA with Shapley-value attribution or partial-information decomposition, enabling principled quantification of redundancy and synergy across information sources.

7. Conclusion

We introduced *Sequential Decomposition and Attribution* (SDA), a model-agnostic evaluation protocol that gates inputs into *Past* (P), *Current* (C), and *Prior* (R) and quantifies their utility, marginal gains, and interactions. Across two datasets (Porto and ETH) and two backbones (LSTM and Transformer), the evidence is consistent: gains from history saturate once a short prefix is available; the most recent state is highly informative at short horizons; and reliable destination priors deliver substantial improvements whose impact grows with horizon length. Interaction analyses indicate structured complementarity: adding C to P yields large reductions at short horizons, and providing R lowers the accuracy versus history curve and reduces the history required to reach saturation. These results argue for input-aware designs that preserve and fuse the last state with high fidelity, exploit goal conditioning when available, and compress or select history beyond the early plateau. Although demonstrated for trajectory forecasting under displacement errors, SDA offers a compact and comparable lens for deciding which inputs matter under which conditions and can be extended to richer priors, objectives, and multi-agent settings.

CRediT authorship contribution statement

Yonghui Liu: Writing – review & editing, Writing – original draft, Visualization, Validation, Methodology, Investigation, Formal analysis, Data curation, Conceptualization; **Inhi Kim:** Writing – review & editing, Supervision, Software, Resources, Project administration, Funding acquisition, Conceptualization.

Data availability

Data will be made available on request.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper

Acknowledgement

This work was supported by the [National Research Foundation of Korea](#) (NRF), Republic of Korea grant funded by the Korea government (MSIT), Republic of Korea (No. RS-2025-02223612).

Appendix A. Multi-seed Overall Performance

[Tables A.6](#) and [A.7](#) report the three-seed ADE/FDE results for all configurations listed in [Table 4](#), providing the robustness analysis referenced in [Section 5.1](#).

Excluding the intentionally low-capacity *Current-MLP*, performance variability is modest across settings. On **Porto**, ADE and FDE exhibit coefficients of variation of 1.2–4.3% and 1.2–2.6%, respectively. On **ETH**, the corresponding ranges are 3.9–10.2% and 3.7–6.2%, reflecting the dataset's smaller size and the higher stochasticity of pedestrian motion. Crucially, the relative ordering among input-source configurations is consistent across all seeds.

These observations justify the single-seed protocol adopted in the main experiments. Given the factorial nature of SDA and the large number of configurations required for history sweeps, prior-quality curves, and interaction studies, the low empirical variance demonstrates that single-seed evaluation offers a reliable accuracy-cost balance for the analyses reported in [Section 4](#).

Table A.6

Three-seed ADE/FDE (mean $\pm$ std) on Porto.

Input Setting	ADE (m)	FDE (m)
Current-MLP	119.34 $\pm$ 2.17	189.00 $\pm$ 3.69
Past-only	104.19 $\pm$ 1.64	150.71 $\pm$ 2.16
Current-only	98.51 $\pm$ 2.73	153.43 $\pm$ 2.41
Past + Current	83.16 $\pm$ 1.02	131.93 $\pm$ 1.55
Current + Prior [†]	90.37 $\pm$ 3.74	133.93 $\pm$ 2.23
Past + Prior [†]	89.11 $\pm$ 2.02	122.08 $\pm$ 3.00
Past + Current + Prior [†]	74.26 $\pm$ 1.40	111.16 $\pm$ 2.34

Table A.7

Three-seed ADE/FDE (mean $\pm$ std) on ETH.

Input Setting	ADE (m)	FDE (m)
Current-MLP	1.338 $\pm$ 0.189	2.216 $\pm$ 0.325
Past-only	0.306 $\pm$ 0.019	0.431 $\pm$ 0.020
Current-only	0.310 $\pm$ 0.024	0.451 $\pm$ 0.021
Past + Current	0.248 $\pm$ 0.017	0.365 $\pm$ 0.013
Current + Prior [†]	0.287 $\pm$ 0.015	0.406 $\pm$ 0.019
Past + Prior [†]	0.283 $\pm$ 0.028	0.376 $\pm$ 0.023
Past + Current + Prior [†]	0.224 $\pm$ 0.009	0.323 $\pm$ 0.012

Appendix B. Additional results on ETH

To complement the Porto experiments in [Section 5.2](#), we report the same history-length sweep on the **ETH** dataset in [Fig. B.11](#). We vary $L \in \{0, 1, \dots, 9\}$ and $H \in \{1, 3, 5, 7, 9, 11\}$ for both backbones.

Compared with Porto, the ETH curves are visibly noisier and less smooth, which is expected given the smaller dataset size and higher stochasticity of pedestrian trajectories. Nonetheless, the qualitative tendency is consistent: the majority of the attainable gain is realized within the first few history frames, after which additional steps contribute little. The coverage-based analysis again identifies an early saturation region (shaded), with L^* around 6–7 for both backbones. Relative-improvement curves fluctuate more than on Porto, but the marginal pattern-steep early improvements followed by a plateau-remains evident.

These results confirm that the observation from Porto generalizes to a different, smaller-scale domain: short histories are typically sufficient, while long histories incur additional computation without proportional accuracy benefit.

Appendix C. General Form of SDA Interaction Metrics

For completeness, we provide the general definition of interaction terms in SDA. While the main text ([Section 3](#)) reports only pairwise and

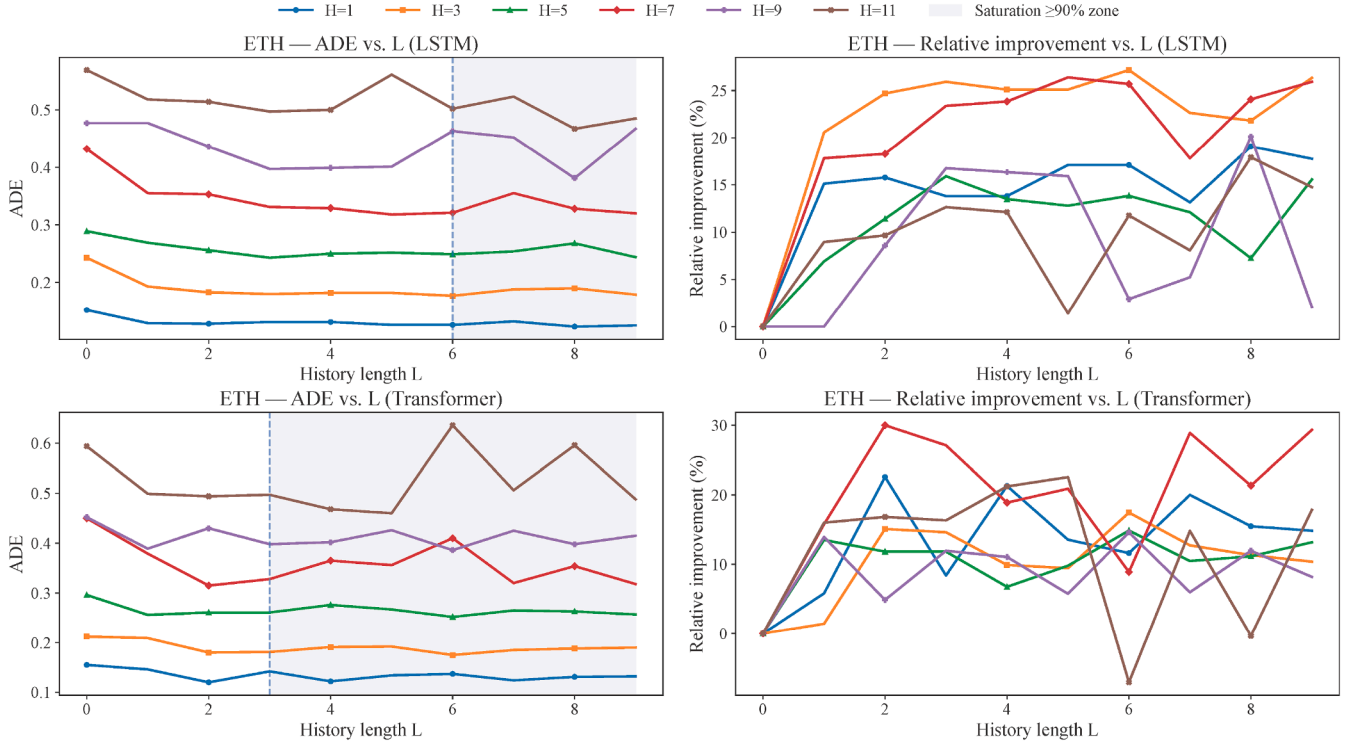


Fig. B.11. ETH - History-length sweep. Left: ADE versus L for LSTM and Transformer at multiple horizons H . Right: relative improvement over $L=0$ (current-only). The shaded band indicates the saturation zone where coverage reaches $p=90\%$ of the total attainable improvement; vertical dashed line marks the corresponding L^* . Compared with Porto, curves are noisier but still show that most gains are achieved by the first few history steps.

conditional interactions for clarity, the formulation extends naturally to arbitrary source subsets.

Let $N = \{P, C, R\}$ denote the full set of sources, and let $A \subseteq N$ be any nonempty subset with $|A| \geq 2$. The **minimum-anchored interaction** for A is defined as

$$I_A^{\min} = M(N \setminus A) - \min_{B \subset A, |B|=|A|-1} M(N \setminus B).$$

A negative value indicates synergy (the whole helps beyond any ($|A| - 1$) subset), while a positive value indicates redundancy. The pairwise interaction in Section 3 is a special case with $|A| = 2$:

$$I_{i,j} = M(N \setminus \{i, j\}) - \min\{M(N \setminus \{i\}), M(N \setminus \{j\})\}.$$

In practice, we restrict our analysis to pairwise and conditional forms (e.g., $I_{R,PC}$, $I_{C,RP}$), since higher-order interactions are difficult to interpret and less stable empirically.

References

- [1] A. Rudenko, L. Palmieri, M. Herman, K.M. Kitani, D.M. Gavrila, K.O. Arras, Human motion trajectory prediction: a survey, *Int. J. Rob. Res.* 39 (8) (2020) 895–935.
- [2] S. Mozaffari, O.Y. Al-Jarrah, M. Dianati, P.A. Jennings, A. Mouzakitis, Deep learning-based vehicle behaviour prediction for autonomous driving applications: A review, *IEEE Trans. Intell. Transp. Syst.* 23 (1) (2022) 33–47.
- [3] A. Alahi, K. Goel, V. Ramanathan, A. Robicquet, L. Fei-Fei, S. Savarese, Social LSTM: human trajectory prediction in crowded spaces, in: *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, IEEE, 2016, pp. 961–971.
- [4] A. Mohamed, K. Qian, M. Elhoseiny, C. Claudel, Social-STGCNN: a social spatio-temporal graph convolutional neural network for human trajectory prediction, in: *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, IEEE, 2020, pp. 14424–14432.
- [5] A. Vaswani, N. Shazeer, N. Parmar, J. Uszkoreit, L. Jones, A.N. Gomez, Ł. Kaiser, I. Polosukhin, Attention is all you need, in: *Advances in Neural Information Processing Systems (NeurIPS)*, 30, 2017, pp. 5998–6008.
- [6] Y. Yuan, X. Weng, Y. Ou, K.M. Kitani, AgentFormer: Agent-aware transformers for socio-temporal multi-agent forecasting, in: *Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV)*, IEEE, 2021.
- [7] K. Mangalam, J. An, H. Girase, J. Malik, It is not the journey but the destination: end-point conditioned trajectory prediction, in: *Proceedings of the European Conference on Computer Vision (ECCV)*, Springer, 2020, pp. 759–776.

- [8] S. Girgis, S. Zagoruyko, A. Alahi, Latentformer: multi-agent transformer for trajectory prediction, in: *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, IEEE, 2022, pp. 9187–9196.
- [9] A. Gupta, J. Johnson, L. Fei-Fei, S. Savarese, A. Alahi, Social GAN: socially acceptable trajectories with generative adversarial networks, in: *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, IEEE, 2018, pp. 2255–2264.
- [10] B. Ivanovic, M. Pavone, Trajectron: dynamic graph generative model for multi-agent trajectory modeling, in: *Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV)*, IEEE, 2019, pp. 2375–2384.
- [11] X. Weng, Y. Yuan, K. Kitani, PTP: Parallelized tracking and prediction with graph neural networks and diversity sampling, *IEEE Rob. Autom. Lett.* 6 (3) (2020) 4640–4647.
- [12] O. Kesa, O. Styles, V. Sanchez, Multiple object tracking and forecasting: Jointly predicting current and future object locations, in: *Proceedings of the IEEE/CVF Winter Conference on Applications of Computer Vision (WACV)*, IEEE, 2022, pp. 560–569.
- [13] S. Casas, B. Aggio, J. Mao, T. Gilles, A. Cui, T. Li, R. Urtasun, Detra: a unified model for object detection and trajectory forecasting, in: *European Conference on Computer Vision*, Springer, 2024, pp. 326–342.
- [14] N. Lee, W. Choi, P. Vernaza, C.B. Choy, P.H.S. Torr, M. Chandraker, DESIRE: Distant future prediction in dynamic scenes with interacting agents, in: *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, IEEE, 2017, pp. 336–345.
- [15] T. Salzmann, B. Ivanovic, P. Chakravarty, M. Pavone, Trajectron++: Dynamically-feasible trajectory forecasting with heterogeneous data, *Proceedings of the European Conference on Computer Vision (ECCV)* (2020) 683–700.
- [16] T. Phan-Minh, H. Do, X. Wang, G. Englebienne, J. Gustafson, A. Alahi, Covernet: multimodal behavior prediction using trajectory sets, in: *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, IEEE, 2020, pp. 14074–14083.
- [17] N. Deo, M. Trivedi, Multimodal trajectory prediction conditioned on lane-graph traversals, in: *Proceedings of the IEEE/CVF Winter Conference on Applications of Computer Vision (WACV)*, IEEE, 2022, pp. 885–894.
- [18] V. Kosaraju, A. Sadeghian, R. Martín-Martín, I. Reid, H. Rezatofighi, S. Savarese, Social-BiGAT: multimodal trajectory forecasting using bicycle-GAN and graph attention networks, in: *Advances in Neural Information Processing Systems (NeurIPS)*, 32, 2019, pp. 137–146.
- [19] K. Mangalam, H. Girase, S. Agarwal, K.-H. Lee, E. Adeli, J. Malik, A. Gaidon, PECNet: Prediction of endpoints conditioned on prior destinations, in: *Proceedings of the European Conference on Computer Vision (ECCV)*, Springer, 2020, pp. 695–712.
- [20] J. Gao, C. Sun, H. Zhao, Y. Shen, D.A. Wang, C. Schmid, C. Li, Vectornet: encoding HD maps and agent dynamics from vectorized representation for motion prediction, in: *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, IEEE, 2020, pp. 11525–11533.

- [21] M. Liang, B. Yang, R. Hu, Y. Chen, S. Liao, S. Feng, R. Urtasun, Learning lane graph representations for motion forecasting, in: *Proceedings of the European Conference on Computer Vision (ECCV)*, Springer, 2020, pp. 541–556.
- [22] F. Giuliari, I. Hasan, M. Cristani, F. Galasso, Transformer networks for trajectory forecasting, in: *Proceedings of the International Conference on Pattern Recognition (ICPR)*, IEEE, 2021, pp. 10335–10342.
- [23] S. Jain, B. Wallace, Attention is not explanation, *Proceedings of the 2019 Conference of the North American Chapter of the Association for Computational Linguistics (NAACL)* (2019) 3543–3556.
- [24] S. Serrano, N.A. Smith, Is attention interpretable?, in: *Proceedings of the 57Th Annual Meeting of the Association for Computational Linguistics (ACL)*, 2019, pp. 2931–2951.
- [25] D. Pruthi, F. Gupta, B. Dhingra, G. Neubig, Z.C. Lipton, Learning to deceive with attention-based explanations, in: *Proceedings of the 58Th Annual Meeting of the Association for Computational Linguistics (ACL)*, 2020, pp. 4782–4793.
- [26] K. Simonyan, A. Vedaldi, A. Zisserman, Deep inside convolutional networks: visualising image classification models and saliency maps, in: *Proceedings of the International Conference on Learning Representations (ICLR)*, 2014, pp.
- [27] M.T. Ribeiro, S. Singh, C. Guestrin, Why should i trust you? explaining the predictions of any classifier, in: *Proceedings of the ACM SIGKDD International Conference on Knowledge Discovery and Data Mining (KDD)*, ACM, 2016, pp. 1135–1144.
- [28] S.M. Lundberg, S.-I. Lee, A unified approach to interpreting model predictions, *Adv. Neural Inf. Process. Syst. (NeurIPS)* 30 (2017) 4765–4774.
- [29] M. Tsang, D. Cheng, H. Liu, Z. Zhang, J. Zhou, Detecting statistical interactions from neural network weights, in: *Advances in Neural Information Processing Systems (NeurIPS)*, 2020.
- [30] Y. Liu, Q. Li, I. Kim, Enhanced trajectory reconstruction from sparse and noisy GPS data: a progressive chunked transformer approach, *Commun. Transp. Res.* 5 (2025) 100200.